

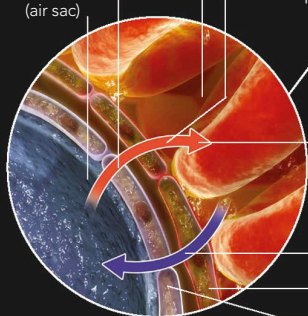
GAS EXCHANGE

THE BODY CANNOT STORE OXYGEN AND NEEDS CONTINUING SUPPLIES. IT ALSO CONSTANTLY PRODUCES CARBON DIOXIDE AS A WASTE PRODUCT. GAS EXCHANGE SWAPS OXYGEN AND CARBON DIOXIDE IN THE LUNGS AND TISSUES.

Oxygen is drawn into the body by the expanding lungs. When it reaches the ends of the lungs' airways, the gas dissolves into the fluid lining the alveoli (air sacs). It passes into the blood for distribution to each body cell. Inside cells, the oxygen reacts with glucose to free its energy. Toxic carbon dioxide is a byproduct of the process, but gas exchange discharges it into the air. In the lungs and body tissues, gases pass by diffusion: flowing from regions of high to low density.

1 Oxygen in air dissolves into fluid lining the alveolus and diffuses through alveolar wall and blood capillary wall.

2 Oxygen enters blood plasma inside capillary



3 Oxygen quickly bonds to haemoglobin in red blood cells

4 Carbon dioxide diffuses out of blood plasma and enters air in alveolus

Cell of capillary wall
Cell of alveolar wall

Deoxygenated blood returns from tissues to heart

Oxygen is drawn into trachea

Heart pumps deoxygenated blood into lungs

Oxygen-rich blood leaves heart

Oxygen-rich blood returns to heart

5 Oxygenated blood leaves heart along the aorta (the body's main artery) and circulates to body tissues

EXCHANGE IN THE LUNGS

When fresh, oxygen-rich air reaches the alveoli – the tiny dead-end air spaces in the lungs – it must pass through several layers to reach the red cells in the blood. But these layers are so thin that the total distance is only 0.001mm ($\frac{1}{2500}$ in).

Lower vena cava (one of the body's two main veins) returns deoxygenated blood from lower body to heart